FINAL YEAR PROJECT PROPOSAL

Face Recognition-Based Attendance System with Liveness Detection

An automated attendance solution using computer vision

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1. Introduction

1.1 Background

Attendance is important in schools, colleges, universities and workplaces. In many places, attendance is still recorded manually by calling names or signing a register. This process can take time, may contain mistakes and can allow another person to mark attendance on someone else's behalf.

1.2 Project Overview

The proposed project is a web-based attendance system that uses face detection and face recognition to identify registered students and record their attendance. A camera captures a face, the system checks it against the registered face data, and attendance is recorded with the date and time when a match is found. A liveness detection feature is also proposed to help identify attempts to use a printed photograph in place of a real person.

1.3 Project Purpose

The purpose is to build a practical student-level application that demonstrates how computer vision and machine learning can support attendance recording. The project will focus on a clear set of features rather than developing a large campus management system.

2. Problem Statement

Manual attendance takes time and depends on the person recording it. It can also lead to incorrect entries, duplicate records or attendance being marked for someone who is not present. Basic camera-based systems may also be vulnerable to simple attempts such as presenting a photograph to the camera. There is a need for a manageable attendance solution that can recognize registered faces, record attendance automatically and test whether a basic liveness check can reduce photo-based misuse.

3. Proposed Solution

This project will develop an application that captures a face through a webcam, detects the face in the frame and compares it with registered face information. When the system identifies a registered student above a selected matching threshold, it will record the student's attendance with a timestamp. If the face is unknown or the match is not reliable, the system will not automatically mark attendance.

A liveness detection component will be explored to help distinguish a live person from a simple printed-photo presentation. The exact method will depend on available time, hardware and test data. The system will store attendance records in a database and provide a basic page for viewing and exporting attendance information. The application will assist the attendance process; its results will still need to be checked during testing.

4. Project Objectives

4.1 General Objective

To design and develop an attendance application that uses face recognition to identify registered students and record attendance, with a focused liveness detection feature to test resistance to simple photo-based spoofing.

4.2 Specific Objectives

- To create a simple interface for registering students and their face images.
- To detect faces from a live camera feed using computer vision.
- To recognize registered students by comparing captured faces with stored face data.

- To record attendance with the student identifier, date and time while preventing duplicate entries for the same session.
- To explore and implement a suitable liveness detection method for basic photo-based presentation attempts.
- To store and display attendance records in a database.
- To evaluate recognition and liveness performance using a labelled test set and different lighting conditions.
- To document the system's limitations, including cases where recognition may fail.

5. Scope of the Project

5.1 In Scope

- Student registration and storage of reference face data.
- Face detection through a webcam or laptop camera.
- Face recognition for registered students.
- Automatic attendance recording with date and time.
- Basic duplicate-attendance prevention for a defined class/session.
- A focused liveness check intended to detect simple printed-photo attempts.
- Attendance history and a basic report view.
- Testing and evaluation of recognition performance.

5.2 Out of Scope

- Complete university management, timetable, fee or course management modules.
- Integration with official university databases or biometric identity systems.
- Guaranteed prevention of every spoofing method, including sophisticated masks or replay attacks.
- Deployment across multiple campuses or large-scale commercial use.
- Use of facial recognition as the only evidence for disciplinary decisions.

5.3 Possible Future Enhancements

If time and supervisor approval allow, future versions could include multi-camera support, more advanced anti-spoofing models, a mobile-friendly interface, timetable integration or notifications for attendance shortages. These features are not required for the initial prototype.

6. Target Users

The main users are a teacher or attendance administrator who registers students and reviews records, and students whose attendance is recorded by the system. The prototype will be designed for a small class or controlled demonstration environment.

7. Proposed Features

7.1 Student Registration

The administrator will enter a student's name or roll number and capture or upload reference face images. Only the minimum information needed for the prototype will be stored.

7.2 Face Detection

The system will locate a face in the camera frame before attempting recognition. Frames without a usable face, or with multiple unclear faces, can be ignored or flagged for another attempt.

7.3 Face Recognition

The system will compare a captured face with registered face data and return a match only when the similarity score meets a chosen threshold. Unknown or low-confidence faces will not be marked present automatically.

7.4 Liveness Detection

A liveness check will be added to help identify simple presentation attacks, such as showing a printed photograph to the camera. The selected approach will be tested against both live participants and photo-based attempts. This feature will be described as a limited protection rather than a guarantee against all spoofing.

7.5 Attendance Recording

When a registered student is recognized and the liveness check passes, the system will save the student's identifier, date, time and session. Duplicate entries for the same session will be prevented.

7.6 Attendance Reports

The administrator will be able to view attendance by date or student and export a simple report if time permits.

7.7 Error Handling

The application will show a clear message when a face is not recognized, the camera is unavailable or the liveness check cannot make a reliable decision.

8. Proposed Technologies

Technology	Planned Use
Python	Main programming language for the application and processing logic.
OpenCV	Camera access, image handling and face-detection support.
Face-recognition model/library	Matching a detected face against registered face data; the final library will be selected after testing.
Flask	A lightweight web application framework for the user interface and routes.
SQLite	Local database for student details and attendance records.
NumPy	Numerical operations and image-data handling where required.
HTML, CSS and Bootstrap	Basic web pages and interface styling.
Git and GitHub	Version control and storage of project code.

The listed technologies are proposed tools. The final stack may be adjusted after an initial feasibility test and supervisor feedback.

9. Expected Outcomes

The expected outcome is a working prototype that can register a small set of students, detect faces from a camera, recognize registered users, record attendance and display attendance history. The prototype will also include a tested liveness-check component intended to reduce simple photo-based attendance attempts.

- A working Python-based attendance application.
- A database containing registered student records and attendance entries.
- A basic liveness detection experiment and documented results.
- Test results showing recognition performance and common failure cases.
- Project source code and documentation maintained in a GitHub repository.

10. Project Significance

This project provides practical experience in computer vision, machine learning, database handling and web application development. It addresses a familiar problem and keeps the main workflow small enough for an individual student project. The evaluation will also help explain the strengths and limitations of face recognition in real conditions, rather than assuming the system is always accurate.

11. Project Feasibility and Ethical Considerations

11.1 Technical Feasibility

The project can be developed using Python and open-source libraries on a standard laptop with a webcam. A pre-trained face-recognition method will be considered to keep the work manageable. The liveness component will be limited to an approach that can be implemented and tested within the project timeline.

11.2 Schedule Feasibility

The project will be developed in stages: first a small camera and face-detection demo, then recognition, attendance storage, liveness testing and final evaluation. This order allows a working core system to be completed before optional improvements.

11.3 Privacy and Ethics

Face images and facial data are sensitive personal information. The prototype should use consented participants or suitable sample data, store only the information needed, and restrict access to the administrator. Face data should not be shared publicly in the GitHub repository. Attendance results may be wrong because of lighting, camera quality or changes in appearance, so the system should provide a way for a teacher or administrator to review and correct records.

11.4 Limitations

Recognition accuracy may change with lighting, camera angle, face occlusion and image quality. A basic liveness check may not detect every photograph, video replay or sophisticated spoofing attempt. The project will report these limitations honestly as part of its evaluation.

12. Development Methodology

An iterative development approach will be used. The system will be built in small modules and tested after each stage. This approach is suitable for an individual project because it makes progress visible and allows problems to be identified early.

1. Requirements and scope: confirm the attendance workflow and agree on the liveness feature with the supervisor.
2. Environment setup: install Python and required libraries, create the GitHub repository and prepare the project structure.
3. Face detection prototype: connect the camera and test face detection under basic conditions.

4. Face registration and recognition: store reference images or face encodings and test matching against registered users.
5. Attendance module: save records with date and time and prevent duplicate entries.
6. Liveness feature: implement a suitable basic check and prepare live-face and photo-attempt test cases.
7. Interface and database: connect the processing modules to the simple web interface and SQLite database.
8. Testing and evaluation: measure recognition results, liveness results and behaviour under different lighting conditions.
9. Documentation and final demonstration: record findings, limitations and instructions for running the prototype.

13. Tentative Project Timeline

The following schedule is a suggested 16-week plan. It can be adjusted according to the university calendar and supervisor feedback.

Weeks	Activity	Expected Deliverable
1–2	Finalize requirements, review related work, set up tools and GitHub.	Approved scope and working environment.
3–4	Build camera access and face-detection prototype.	Demonstration of face detection.
5–6	Register student samples and implement face recognition.	Initial recognition module.
7–8	Build attendance storage and duplicate-entry checks.	Attendance records saved in SQLite.
9–10	Implement the simple liveness-check feature.	Liveness prototype and test cases.
11–12	Connect modules to a basic Flask interface and reports.	Integrated web prototype.
13–14	Run tests with different people, lighting and photo attempts.	Evaluation results and identified limitations.
15	Fix issues and complete documentation.	Final report and user instructions.
16	Prepare presentation and demonstration.	Final FYP submission.

14. Evaluation Plan

The system will be evaluated using a small, clearly labelled test set. Testing should include registered participants under different lighting conditions, unknown participants and simple photo-based attempts. Results will be recorded rather than assumed.

- Face recognition: count correct identifications, incorrect identifications and missed identifications.
- Liveness detection: record how often live participants are accepted and photo attempts are rejected.
- Attendance logic: check timestamp recording, duplicate prevention and correct student-to-record mapping.
- Usability: ask a small number of test users to complete registration and attendance tasks and note any problems.

Possible metrics include recognition accuracy, precision, recall and the false acceptance / false rejection rates, where the test set is large and balanced enough to support them. The final report will state the size and limits of the test data.

15. References

- OpenCV. (n.d.). OpenCV Documentation. https://docs.opencv.org/
- Python Software Foundation. (n.d.). Python Documentation. https://docs.python.org/3/
- Flask. (n.d.). Flask Documentation. https://flask.palletsprojects.com/
- scikit-learn developers. (n.d.). scikit-learn User Guide. https://scikit-learn.org/stable/user_guide.html
- SQLite. (n.d.). SQLite Documentation. https://www.sqlite.org/docs.html
- GitHub. (n.d.). GitHub Docs. https://docs.github.com/

Note: Before final submission, add the exact research papers or library documentation consulted during implementation, using the reference style required by the department.

16. Conclusion

The proposed Face Recognition-Based Attendance System with Liveness Detection aims to make attendance recording quicker and more consistent by recognizing registered students through a camera. It combines a focused computer-vision workflow with attendance storage and a basic attempt to detect photo-based misuse. The scope is designed for an individual IT student and avoids building a complete management system. Through testing and documentation, the project will demonstrate both the practical value and the limitations of automated face-based attendance.

End of Proposal